

Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

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Structure-preserving representation before decision contraction.

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pip install erisml-compiler · github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc

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Don't collapse to a scalar.

	harm	rights	consent	risk	fair
speaker	red	orange	purple	orange	red
village	red	red	orange	red	orange
refugees	green	green	green	orange	green
nazis	orange	purple	orange	orange	purple

the moral tensor (who x what is owed)

✓ keep the structure

who is affected · what is owed
who consented · who bears imposed risk

contract to a decision only at the end,
and log the contraction:
weights · who lost · what residue remains

★ 0.42
one scalar — structure lost

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Four lenses. No silent winner.

one MoralGraph

Consequentialist harm/care tensor
Gini · worst-off · Shapley

Deontic (Kantian) gates · universalizability via Z3 SMT

Virtue Aristotelian habit-consistency

Care ethics Gilligan · Noddings · Tronto

verdicts disagree? → cross_projection_disagreement

defers to a human

"We run four and show you where they conflict — we don't pick one and hide it."

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Safety in the loop. Fails safe.

Agent planner

Reflex < 100 μs · hard stops

Tactical ErisML · 10-100 ms

Strategic policy + human oversight

ALLOW REVISE BLOCK

proof proof proof proof

SHA-256 DecisionProof chain
previous_proof_hash → proof_hash

ethics service times out → rule-based fallback. Never fails open.

between the agent's planner and its actuators

2

pip install erisml-compiler — structure before the scalar

text → segment → extract → canonicalize → MoralGraph → tensorize → DEME → DecisionProof

MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm — carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable) · LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

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Is this 'good'? → Does it preserve the bonds?

Obligation O Claim C

Liberty L No-claim N

s = correlative swap (agent ↔ patient)

r² = deontic negation

s and r² commute → V4 (Klein four-group, order 4) — measured

full D4 (order 8) posited — quarter-turns not yet observed

Bond Index: does the judgment survive the swap?

0.0 perfect 0.155 baseline 0.25 warn 0.30 block

multi-rank tensors: party x time x action x coalition

V4 claim machine-checked: Lean 4 + Mathlib (formal/HohfeldV4.lean)

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What the model says vs what it exhibits

transformer (Qwen2.5-7B, LLaMA, Mistral)

text lens (what it says)

activation lens (what it exhibits)

delta lens compare

requires human review (5 failure modes)

activation / probe lens is EARLY — uncalibrated by default (research-grade)

turboquant-pro (PyPI): PyTorch-native compression
HF 1-liner: TurboQuantic-ai-hel · Triton + Volta kernels · vLLM -5x KV
live here: 3-bit embedding codec on NATS · claims CI-gated (CLAIMS.md)

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Auditable means you can replay the judgment

stakeholder	harm	verdict
speaker	0.76	forbid
village	0.83	forbid
refugees	0.00	prefer
nazis	0.18	neutral

Gini(harm) = 0.43 · worst-off = village

Shapley: speaker 7.11 · refugees 7.70
nazis 7.88 · village 7.18

DecisionProof: proof_hash → audit.ir_hash

worked example examples/nazi_attic · rank-2 DEME — one command, real numbers, a hash you can verify.

One command. Real numbers. A hash you can verify.

\$ erisml-compile examples/nazi_attic

per-party verdicts · Gini 0.43 · exact Shapley
DecisionProof chained to the IR hash — replay the judgment

1. Structure before contraction — keep the tensor, log the collapse + the residue.
2. Pluralism is the responsible move — four lenses, defer to a human on conflict.
3. Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early / uncalibrated
ranks 1-3 real, higher partial · V4 measured, D4 posited (Lean + Mathlib verified)
embodied = design target · conversational agents = live

